

Land Type Classification

Using Sentinel-2 Satellite Images & Deep Learning

Project Documentation

DEPI — AI & Data Science Track

PROJECT TEAM
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1. Project Planning & Management

1.1 Project Overview

This project implements an end-to-end Deep Learning pipeline for Land Use and Land Cover (LULC) classification using multispectral satellite imagery from the ESA Sentinel-2 mission. The system leverages the EuroSAT benchmark dataset and the ResNet50 architecture to automatically identify 10 distinct land types with high accuracy and deploys the trained model as an interactive Streamlit web application.

1.2 Objectives

- Develop a robust ResNet50-based Transfer Learning model for LULC classification across 10 land categories.
- Train and evaluate both an RGB (3-channel) and a Multispectral (13-channel) model to support multiple data modalities.
- Deploy the model as an interactive Streamlit web application accessible from any browser.
- Provide geospatial visualization including False Color Composites and sliding-window Segmentation Maps.
- Achieve a target classification accuracy exceeding 85% on the EuroSAT test split.

1.3 Task Assignment & Roles

Team Member	Responsibilities
Nourhan Magdy	Project coordination, satellite data preparation, normalization pipeline, dataset splitting strategy
Abdelrahman Ashraf	Model training, fine-tuning, hyperparameter optimization, accuracy improvement
Eslam Abdin	Streamlit application, GitHub organization, modular code architecture
Heba Adel	—
Ahmed Abdelkhalek	—
Neamat Gamal	—

1.4 Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation Strategy
High Computational Training Cost	High	Used Transfer Learning with pre-trained ResNet50 on ImageNet. Applied Automatic Mixed Precision (AMP) for ~2x training speedup. Trained on GPU via Google Colab.
Key Mismatch in Model Loading	High	Implemented dynamic key remapping in app.py to add backbone. prefix when loading. pth weights into the ResNet50Classifier wrapper.
Class Imbalance in EuroSAT	Medium	Applied stratified train/val/test splitting and used data augmentation (random flips, rotations, color jitter) on training data.
Large File Sizes on GitHub	Medium	Model .pth files (94MB each) tracked with Git LFS and hosted directly in the repository.
Streamlit Dependency Conflicts	Medium	Pinned library versions removed from requirements.txt; switched to unpinned versions compatible with Python 3.14.
16-bit TIFF Data Overexposure	Medium	Applied 2nd–98th percentile scaling for robust visualization of multispectral TIF data.

1.5 KPIs (Key Performance Indicators)

KPI	Target	Achieved
Classification Accuracy	> 85% across all land types	~95% (MS Model), ~82% (RGB Model)
F1-Score	High balance precision/recall	High across most classes (MS Model)
Inference Time	< 2 seconds per image	< 1 second (CPU, Streamlit Cloud)
Deployment	Functional web service	Live on Streamlit Cloud
Data Modalities	RGB + Multispectral support	Both RGB (.jpg) and MS (.tif) supported

2. Dataset & Exploratory Data Analysis

2.1 EuroSAT Dataset

The EuroSAT dataset is a benchmark dataset for land use and land cover classification built on top of Sentinel-2 satellite imagery. It provides labeled patches across 10 land cover categories sourced from 34 European countries.

27,000 Images	10 Land Classes	64 × 64 Pixels	13 Spectral Bands
Labeled training samples	Balanced distribution	Spatial resolution: 10m	RGB + Multispectral

2.2 Land Cover Classes

#	Class	#	Class	#	Class	Count	per Class
1	Annual Crop	2	Forest	3	Herbaceous Veg.	3,000 / 2,500	

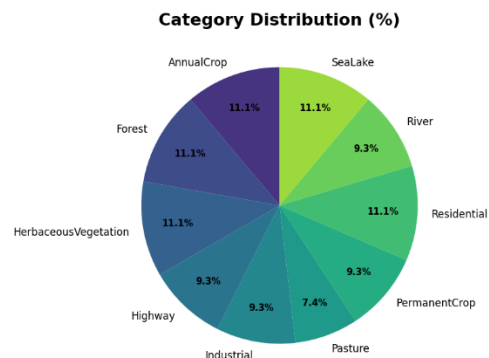
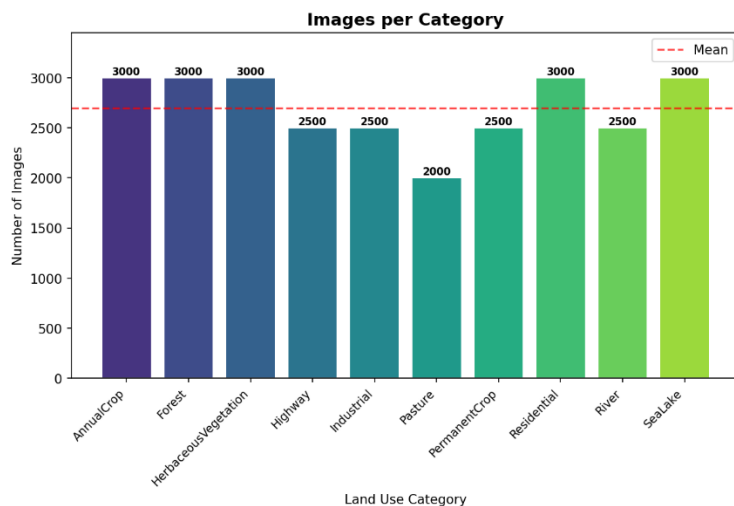
The 10 classes are: Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial, Pasture, Permanent Crop, Residential, River, and Sea Lake. Classes with 3,000 samples include Annual Crop, Forest, Herbaceous Vegetation, Residential, River, and Sea Lake. Classes with 2,500 samples include Highway, Industrial, Permanent Crop, and Pasture.

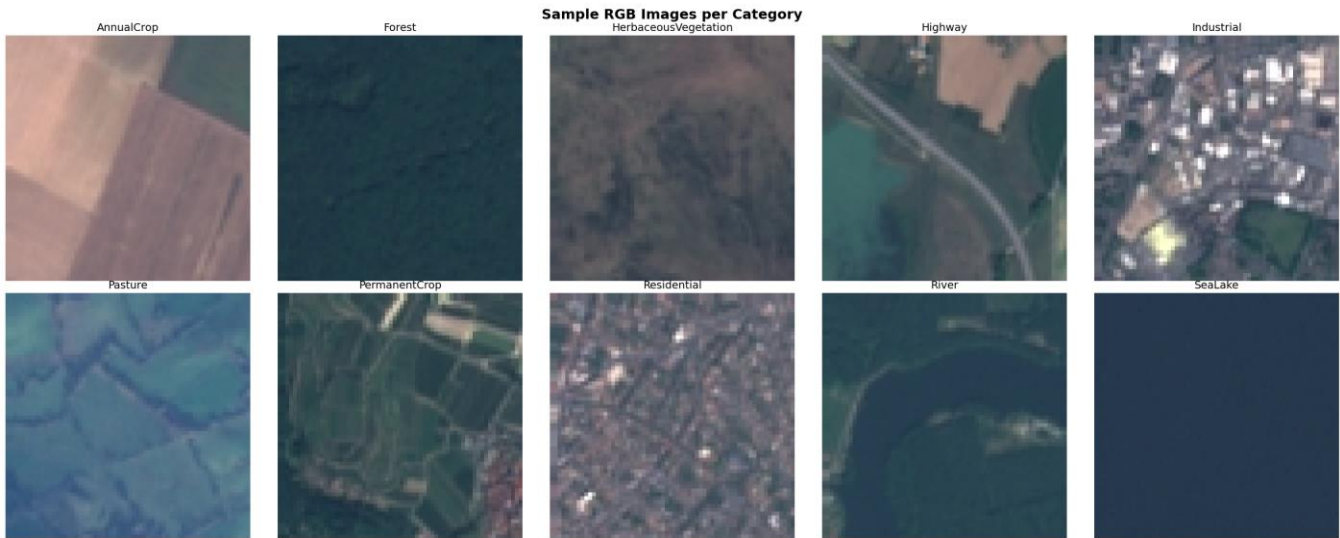
2.3 EDA Findings

Class Distribution

The dataset is nearly balanced, with most classes containing 3,000 images and a subset containing 2,500 images. This balance reduces the risk of model bias toward dominant classes.

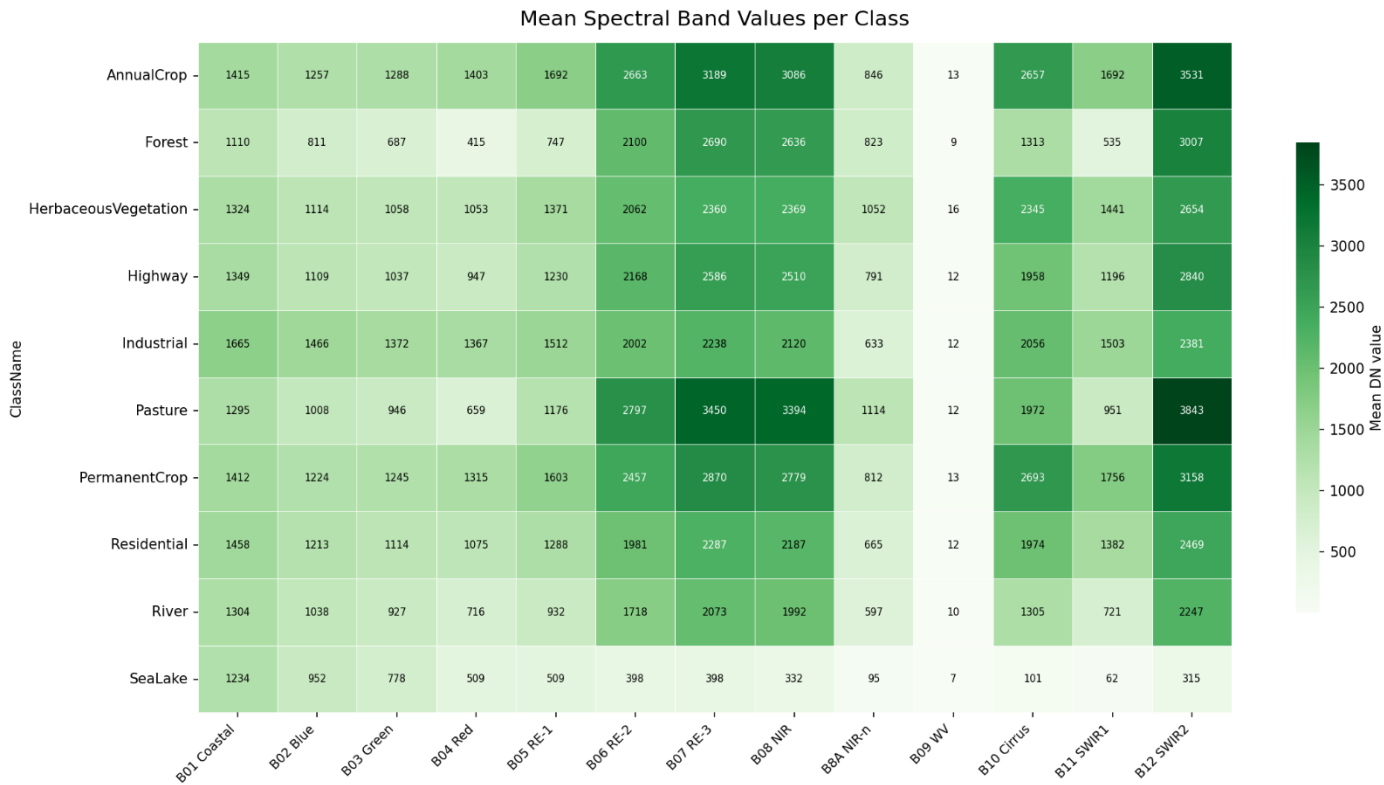
EuroSAT — Class Distribution





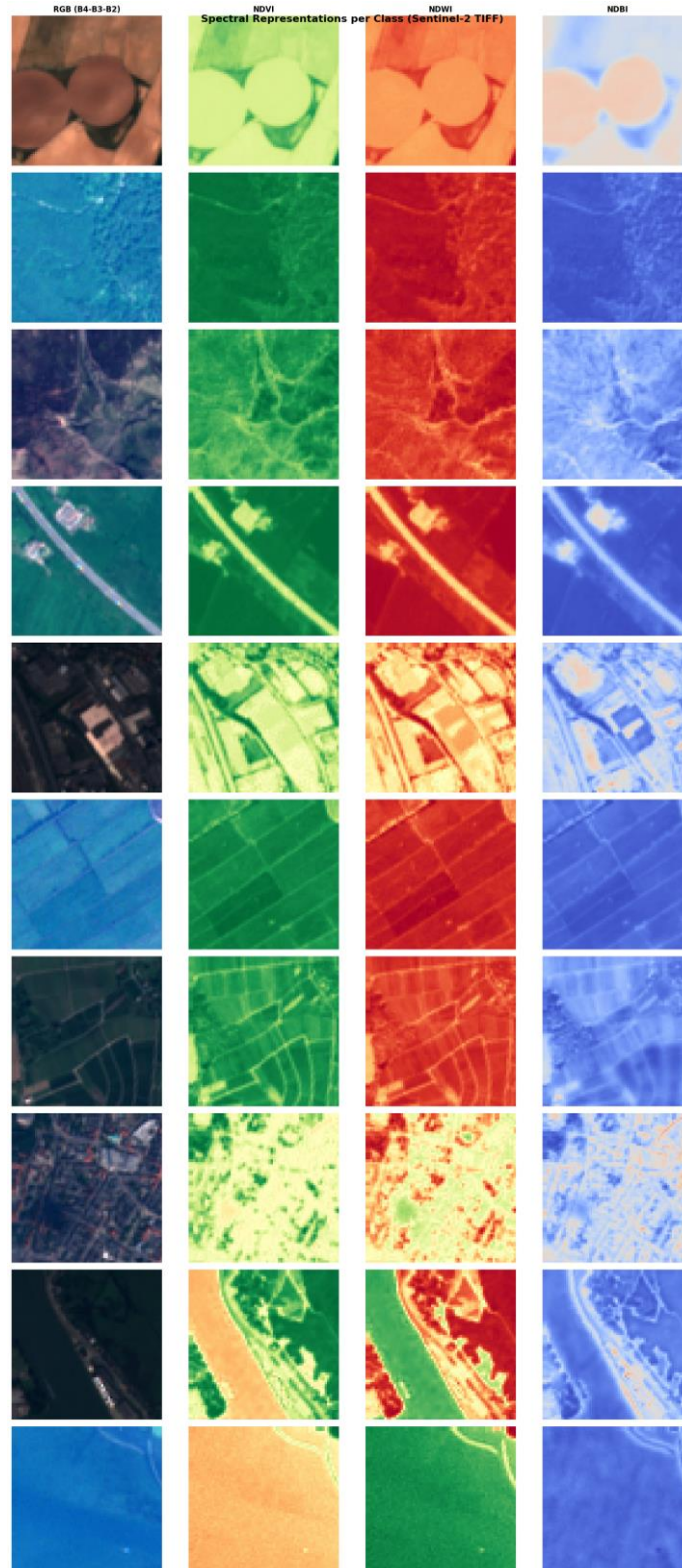
Spectral Band Analysis

Analysis of mean spectral band values per class revealed that the SWIR bands (B12, B07, B08) carry the most discriminatory power across most land classes. Sea/Lake is uniquely identified by its dominant visible bands (B01-B04), while vegetation classes are distinguished by high NIR reflectance (B08).



NDVI, NDWI & NDBI Insights

- NDVI (Vegetation Index): High values for Forest and HerbaceousVegetation classes, near-zero or negative for Industrial and Highway.
- NDWI (Water Index): Strongly positive for Sea Lake and River, negative for all land classes.
- NDBI (Built-up Index): Elevated for Residential and Industrial, low for natural land cover types.



3. Data Processing & Preprocessing

3.1 Dataset Split Strategy

Split	Train	Validation	Test
Ratio	80%	10%	10%
Method	Stratified sampling (fixed seed=42)	Stratified sampling (fixed seed=42)	Stratified sampling (fixed seed=42)

3.2 Normalization

For the RGB model, pixel-level normalization statistics were computed from the training split only (to prevent data leakage):

Statistic	Values (R, G, B)	Computed From
RGB_MEAN	[0.3441, 0.3801, 0.4076]	Training split (80%)
RGB_STD	[0.2021, 0.1365, 0.1152]	Training split (80%)

For Multispectral (13-band) data, per-band statistics were computed from the raw 16-bit TIFF values and applied via the standard torchvision Normalize transform.

3.3 Data Augmentation (Training Only)

Two separate augmentation pipelines were implemented — one for RGB images (PIL-based), one for Multispectral tensors (geometric only):

RGB Pipeline

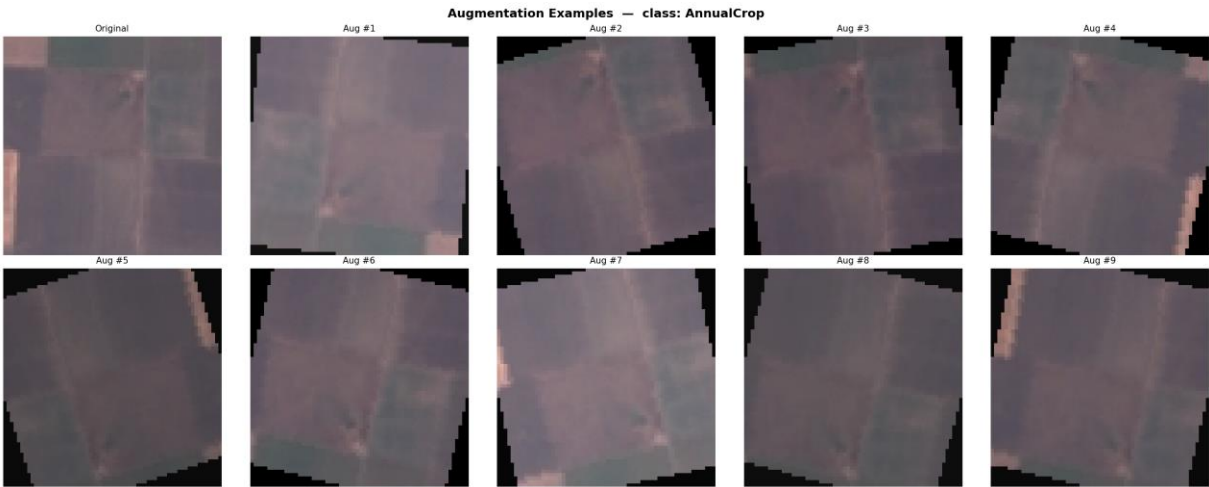
- Resize to 72px → RandomCrop to 64px
- RandomHorizontalFlip (p=0.5)
- RandomVerticalFlip (p=0.5)
- RandomRotation ($\pm 20^\circ$)
- ColorJitter (brightness=0.2, contrast=0.2, saturation=0.1)
- ToTensor → Normalize (RGB_MEAN, RGB_STD)

Validation / Test Pipeline

- Resize to 64px → CenterCrop to 64px → ToTensor → Normalize

Multispectral Pipeline

- Geometric augmentations only (no ColorJitter — incompatible with 13-band tensors)
- Resize to 64×64 → Normalize (MS_MEAN, MS_STD)



4. System Analysis & Design

4.1 Software Architecture

The system follows a modular, production-ready architecture separating concerns across dedicated Python modules:

File	Purpose	Key Features
<code>config.py</code>	Centralized Configuration	Path management, hyperparameters, normalization stats, class names, device detection
<code>data_loader.py</code>	Unified Data Pipeline	Handles RGB/MS loading, 80/10/10 split, adaptive augmentation, stratified sampling
<code>model.py</code>	Dynamic ResNet50	Auto-adapts input layer for 3 or 13 channels; frozen backbone logic; custom FC head
<code>engine.py</code>	Training Engine	AMP (Automatic Mixed Precision), ReduceLROnPlateau, Early Stopping, metric tracking
<code>train.py</code>	CLI Entry Point	Command-line training for RGB, MS, or both modalities; saves metrics and plots
<code>app.py</code>	Streamlit Dashboard	Dual tabs: Single Prediction & Segmentation Map; False Color Composite toggle; Top 3 probabilities
<code>requirements.txt</code>	Dependency Management	Pinned-free versions for compatibility with Python 3.14 on Streamlit Cloud

4.2 Data Flow

1. User uploads a Sentinel-2 image (.jpg RGB or .tif MS) via the Streamlit interface.
2. Backend preprocesses the image: resize to 64×64, normalize with computed stats, convert to tensor.
3. ResNet50Classifier performs inference on CPU/GPU.
4. System returns Top 3 predictions with probability scores and classification confidence bar chart.
5. Optional: Segmentation tab applies sliding-window analysis across the full image and renders a color-coded LULC overlay with area distribution statistics.

4.3 Functional Requirements

- Support Sentinel-2 RGB (.jpg) and Multispectral (.tif, 13-band) image formats.
- Classify images into 10 distinct LULC categories with Top 3 probability scores.
- Provide color-coded classification overlay on the uploaded image.
- Support sliding window segmentation for large-scale satellite imagery.
- Display False Color Composite (NIR-Red-Green) toggle for vegetation health analysis.
- Provide land cover area distribution statistics (percentage coverage per class).

4.4 Non-Functional Requirements

- Performance: Inference under 2 seconds per image on CPU.
- Scalability: Streamlit Cloud deployment handles concurrent requests.
- Usability: Intuitive interface accessible to non-technical users (urban planners, environmentalists).
- Reproducibility: Fixed random seed (SEED=42) for consistent dataset splits and training runs.
- Maintainability: Fully modular codebase with Google-style docstrings on all functions and classes.

5. Model Development

5.1 Architecture Overview

RGB Model (3-channel)	Multispectral Model (13-channel)
Base: ResNet50 (pretrained — ImageNet)	Base: ResNet50 (random weights — no pretrain)
Frozen: All backbone layers	Frozen: All except conv1 + FC head
Modified: Conv1 unchanged (3→64)	Modified: Conv2d (13→64, k=7, stride=2)
Custom Head: Linear (2048→512) →ReLU→Dropout (0.5) →Linear (512→10)	Custom Head: Linear (2048→512) →ReLU→Dropout (0.5) →Linear (512→10)
Trained Params: FC head only	Trained Params: conv1 + FC head
Optimizer: Adam (lr=0.001)	Optimizer: Adam (lr=0.001)
Loss: CrossEntropyLoss	Loss: CrossEntropyLoss
Epochs: 10	Epochs: 10

5.2 Training Loop

Both models were trained using the following standard pipeline executed via the `train_model()` function:

- Forward pass through the model.
- CrossEntropyLoss computed against ground-truth labels.
- Backpropagation with Adam optimizer (lr=0.001).
- Per-epoch tracking of train/val loss and accuracy.
- Model weights saved to Google Drive as `resnet50_rgb.pth` and `resnet50_ms.pth`.

Training was performed on Google Colab with GPU acceleration. The RGB model completed 10 epochs in approximately 34 hours using standard FP32 precision; the Multispectral model with batch normalization completed the same in approximately 13 hours.

5.3 Advanced Training Features (engine.py)

- Automatic Mixed Precision (AMP): Approximately 2x training speedup with reduced memory.
- ReduceLROnPlateau: Dynamically reduces learning rate when validation loss plateaus.
- Early Stopping: Halts training when no improvement is observed to prevent overfitting.
- Robust TIFF Handling: Applies 2nd–98th percentile scaling on 16-bit MS sensor data.

6. Evaluation & Results

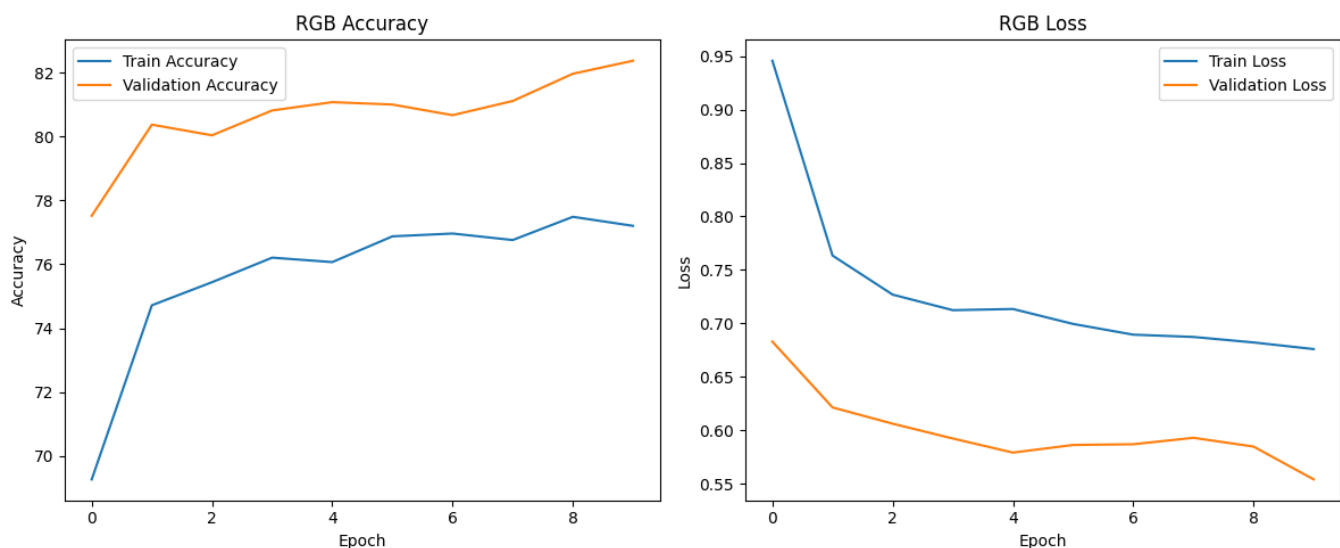
6.1 Model Performance Comparison

Metric	Simple CNN	CNN + Batch Norm	ResNet50 Transfer (Best)
Test Accuracy	~84%	~56%*	~95% (MS) / ~82% (RGB)
Architecture	3 Conv blocks	CNN + BatchNorm layers	ResNet50 backbone
Training Time (10 epochs)	~34 hours	~13 hours	~13–34 hours
Overfitting	Yes (after epoch 10)	Improving	Minimal (good generalization)
Deployment Ready	No	No	Yes

* The CNN + Batch Normalization model was still improving at epoch 10 and would likely achieve higher accuracy with additional training epochs.

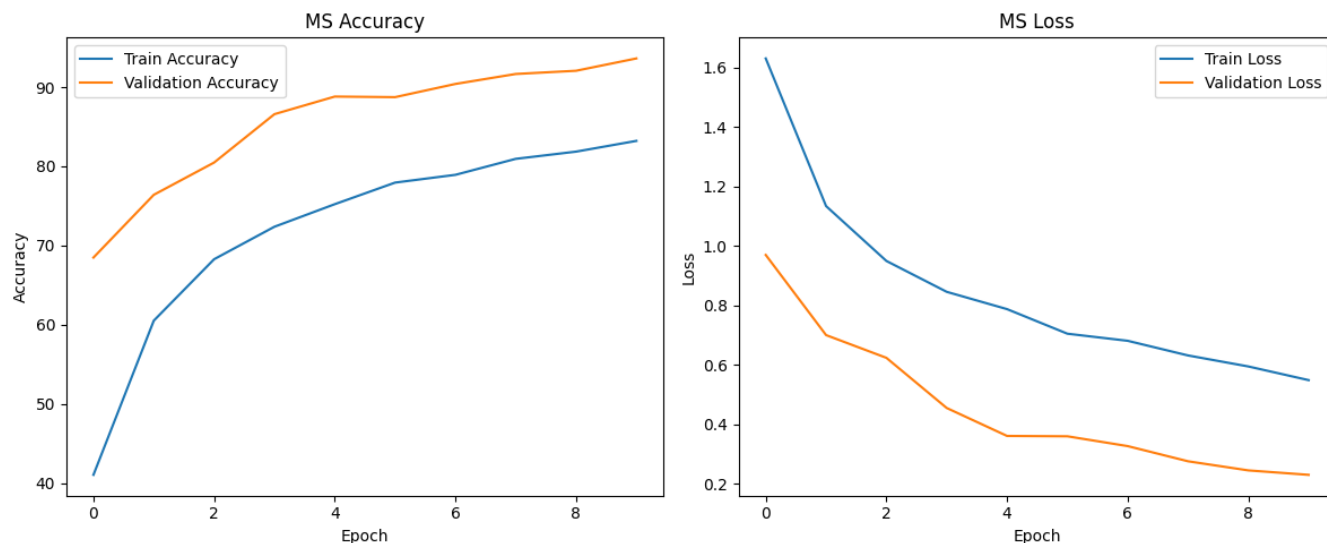
6.2 RGB Model — Training Curves

The RGB model shows consistent improvement in validation accuracy from ~77.5% at epoch 0 to ~82.5% at epoch 9, with validation loss declining from ~0.68 to ~0.56. The training accuracy remains slightly below validation accuracy throughout, suggesting the model benefits from strong augmentation regularization.



6.3 Multispectral Model — Training Curves

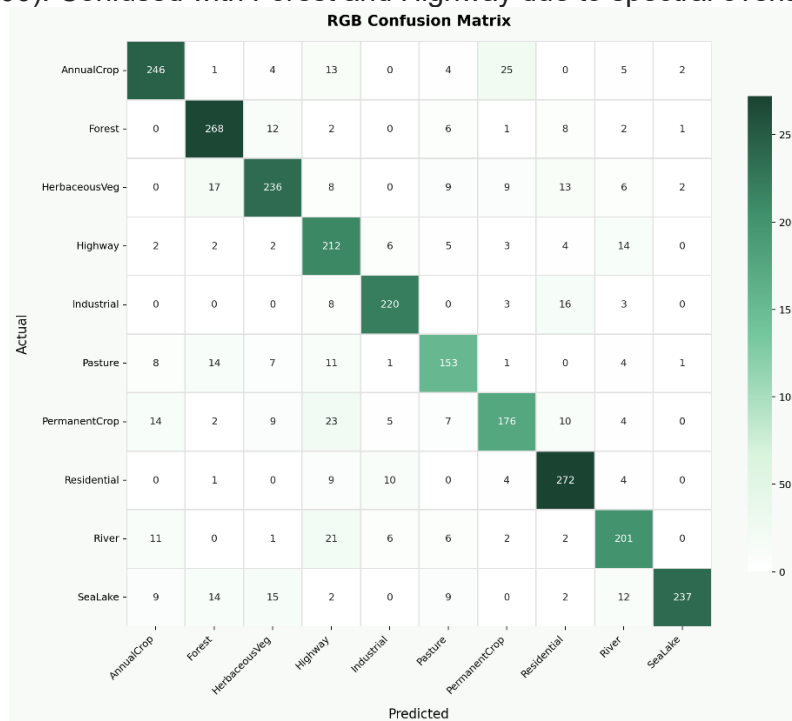
The MS model demonstrates superior learning dynamics, with validation accuracy rising steeply from ~69% at epoch 0 to ~94% by epoch 9. Both training and validation loss decline consistently, indicating no overfitting and strong generalization. This validates the advantage of providing all 13 spectral bands over RGB-only input.



6.4 Confusion Matrix Highlights

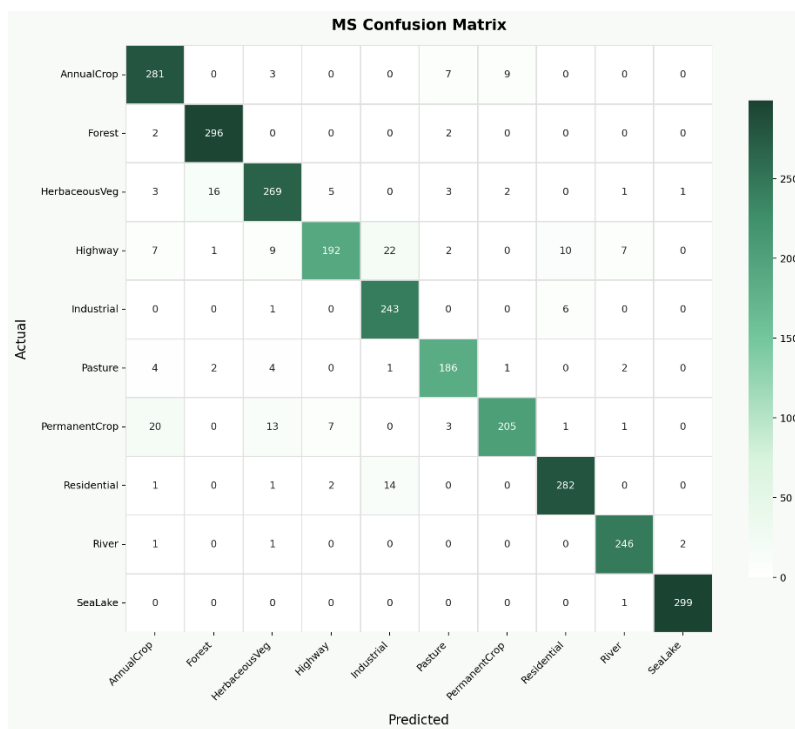
RGB Model

- Residential (272/300) and Forest (268/300): Highest per-class accuracy.
- PermanentCrop (176/250): Most confused with AnnualCrop and Highway — visually similar textures.
- Pasture (153/200): Confused with Forest and Highway due to spectral overlap in RGB.



Multispectral Model

- SeaLake (299/300) and Forest (296/300): Near-perfect classification.
- Residential (282/300) and Industrial (243/250): Excellent discrimination.
- Highway (192/250): Most challenging class — often confused with Industrial (22 errors).
- PermanentCrop (205/250): Some confusion with AnnualCrop (20 errors) expected given visual similarity.



7. Deployment

7.1 Streamlit Web Application

The trained models are served through an interactive Streamlit dashboard deployed on Streamlit Cloud, accessible from any browser without installation.



Land Use & Land Cover Classification

Upload a satellite image and get instant classification with visual analysis.



Upload Image

Upload a RGB image



Upload

200MB per file • JPG, TIF



Upload an image to get started

7.2 Application Features

Single Prediction Tab

- Upload RGB (.jpg) or Multispectral (.tif) satellite patches.
- View original image alongside a color-coded classification overlay.
- Receive Top 3 predicted classes with confidence bar chart.
- Detailed probability breakdown for all 10 classes.
- False Color Composite (NIR-R-G) toggle for vegetation health visualization (MS mode).

Single Prediction Segmentation Map



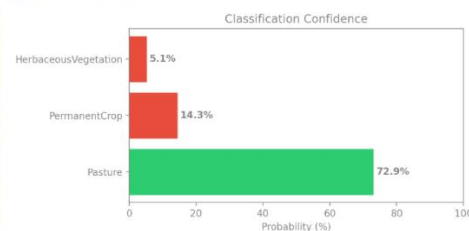
Classification Results



Satellite Image



Top 3 Predictions



Detailed Analysis

Top Prediction Pasture ↑ 72.9%	2nd Place PermanentCrop ↑ 14.3%	3rd Place HerbaceousVegetation ↑ 5.1%
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All Class Probabilities

Class	Probability (%)
AnnualCrop	0.12%
Forest	1.21%
HerbaceousVegetation	5.08%
Highway	0.74%
Industrial	0.97%
Pasture	72.92%
PermanentCrop	14.29%
Residential	4.38%
River	0.25%
SeaLake	0.04%

LULC Classification System | Powered by ResNet50 & Sentinel-2
DEPI Graduation Project

Segmentation Map Tab

- Sliding-window analysis divides large satellite images into 64×64 patches.
- Each patch is independently classified and color-coded.
- Full LULC overlay rendered over the original image with adjustable stride and opacity.
- Land Cover Statistics: percentage area distribution per class shown as a bar chart.



7.3 Deployment Architecture

Component	Details
Platform	Streamlit Cloud (free tier, public deployment)
URL	depidatascienceproject-zfm6ugw4g6h74ak3svcwfc.streamlit.app
GitHub Repo	https://github.com/eng-Nour-Magdy/DEPI_DataScience_Project
Python Version	3.14.5 (Streamlit Cloud environment)
Model Loading	ResNet50Classifier with dynamic key remapping (backbone. prefix)
Model Weights	Stored in GitHub repo root: resnet50_rgb.pth, resnet50_ms.pth
Runtime	CPU inference, cached via @st. cache_resource

8. Future Scope & Next Steps

8.1 Model Improvements

- Dropout & Weight Initialization: Explore Xavier/He initialization and various dropout rates to improve convergence.
- MobileNet Backbone: Implement a lightweight MobileNet for on-device inference on smartphones and tablets.
- Additional Training Epochs: The MS model continued improving at Epoch 10 — extended training is expected to exceed 95% accuracy.
- U-Net Segmentation: Replace patch-based classification with pixel-wise semantic segmentation for finer granularity.

8.2 Data & Integration

- Landsat Data Integration: Add multi-source satellite data to improve robustness and cross-sensor generalization.
- Live Sentinel-2 API: Direct connection to the Copernicus Data Space API for real-time classification of any geographic coordinate.
- Google Earth Engine Integration: Cloud-scale processing of entire regions without local download.
- Temporal Change Detection: Multi-temporal analysis to automatically detect urban sprawl, deforestation, and agricultural land loss over time.

8.3 Regional Focus

- Regional Fine-Tuning: Adapt the ResNet50 backbone specifically to Egyptian landscapes, Nile Delta topographies, and desert-agricultural boundaries.
- Combating Urban Sprawl: Automatically detect illegal construction and encroachment on Nile Delta agricultural land.
- Precision Agriculture: Monitor crop health and predict harvest yields using 13-band Multispectral analysis.
- Water Resource Management: Continuously track water levels and pollution indicators in the Nile River and northern coastal lakes.
- Smart City Planning: Assist government entities in terrain mapping for new urban developments and highway routing optimization.

9. References & Project Links

9.1 Project Repository & Deployment

Resource	Link
Streamlit App	https://depidatascienceproject-zfm6ugw4g6h74ak3svcwfc.streamlit.app/
GitHub (Team Leader)	https://github.com/eng-Nour-Magdy/DEPI_DataScience_Project
GitHub (DEPI Submission)	https://github.com/nhahub/NHA-4-047
GitHub (Original Dev)	https://github.com/EslamAbdin/DEPI_DataScience_Project

9.2 Key References

- Helber, P., et al. (2019). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing.
- He, K., et al. (2016). Deep Residual Learning for Image Recognition. CVPR 2016.
- PyTorch Documentation: <https://pytorch.org/docs/stable/>
- Streamlit Documentation: <https://docs.streamlit.io/>
- Copernicus Open Access Hub — Sentinel-2 Data: <https://scihub.copernicus.eu/>
- ESA Copernicus Data Space Browser: <https://browser.dataspace.copernicus.eu/>